INCH-POUND

MIL-C-48868E (AR)
30 March 1990
SUPERSEDING
MIL-C-48868D (AR)
7 May 1982

MILITARY SPECIFICATION

CHARGE, PROPELLING, M204
PARTS AND CONTAINER ASSEMBLY

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and packaging for the parts and container assembly for the propelling charge designated as Charge, Propelling, M204 manufactured by the slurry process and used on the Lightweight Company Mortar System (LWCMS).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and Supplement there to, cited in the solication (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General. Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	– Sampling procedures and Tables for Inspection by Attributes
MIL-STD-286	- Propellants, Solid: Sampling, Examination and Testing

(Unless otherwise indicated, copies of Federal and Military specifications, standards, and handbooks are available from the Military Specifications and Standards, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other government documents, drawings, and publications. The following other government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (See 6.6)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9277371	- Composition (Closure)
9312697	- Container Assembly
9312698	- Charge, Propelling M204
9313703	- Packing and Marking for BOX, Fiberboard, Outer for Charge propelling, M204 or Container Assembly
9313727	- packing for Pallet for Box, Fiberboard, outer for Charge Propelling M204 or Container Assembly
9381562	- Composition (Container)

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the reference cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Materials. The materials including packing shall be in accordance with the applicable drawings and specifications .

3.2 Assemblies. Container assemblies shall comply with all the requirements specified on drawing 9312697 and on all related drawings and with all the requirements specified in the applicable specification.

3.3 Stability heat test (134.5°C). The nitrocellulose in the container assemblies and closure shall not completely change the color of the methyl violet paper to salmon pink color in less than thirty-five minutes when tested in accordance with 4.5.2.

3.4 Percent by weight of resin and additives, kraft and polyester staple fiber. The percent by weight of resin and additives, kraft and polyester, staple fiber for the container assembly shall not be less than 12.95 and not more than 21.55 when tested in accordance with 4.5.1.3.5. The percent by weight of resin and additives, kraft and polyester, staple fiber for the closure shall not be less than 19.95 and not more than 28.55 when tested in accordance with 4.5.1.3.5.

3.5 Residue test. The container assemblies when loaded and fired ballistically shall not leave residue which will cause cartridge hang-up in the mortar tube, shall not produce a secondary flame after 2 seconds (sec.) minimum (min.) from the primary flame of the cartridge launch and when fired at zone 4, shall not produce flaming debris on the ground. The test shall be performed as specified in 4.5.3.

3.6 Compression strength test. The container assemblies shall. exhibit a compress-ion strength of not less than 0.9 pounds when tested as specified in paragraph 4.5.4.

3.7 Fin-boom simulation test. The container assemblies shall satisfactorily pass the fin-boom simulation test as specified in 4.5.5.

3.8 Water repellancy. The container assemblies shall satisfactorily pass the water repellancy test specified in 4.5.6.

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3.9 Homogeneity of material. The material used to manufacture the parts and assemblies shall be uniform in structure or composition throughout. The presence of spots of 1/16 inch or larger diameter or any spot which completely permeates through the entire thickness of the part or assembly shall indicate non-homogeneity . The presence of thin membrance-like areas shall indicate non-homogeneity. The top and bottom halves shall not exhibit non-homogeneities including cracks, splits, cuts, tears, rips or holes when tested as specified in 4.5.7. The Government shall establish visual standards prior to production with the assistance of the contractor.

3.10 Process description. A process control document, including a complete process description in accordance with DI-P-1604 (Modified) is required. Government review of the process control document is required prior to initiation of production.

3.11 First article inspection. This specification contains technical provisions for first article inspection. Requirements for submission of first article samples by the contractor shall be as specified in the contract.

3.12 Workmanship.

3.12.1 Container parts and assemblies. The container parts and assemblies shall be free of dirt, grease, cracks, splits, cuts, tears, rips, holes, soft spots, seam separations or foreign materials.

3.12.2 Packaging. The packaging shall be free from foreign material. The boxes shall be assembled without undue binding and no flap shall project beyond the edge of the box when closed .

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and

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services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detailed specification.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government fox acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The following types of inspections shall be conducted on this item:

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting officer for valuation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated:

Part Description	Drawing	Quantity
Container, Top	9312699	15 per cavity/(350)
Container, Bottom	9312700	15 per cavity/(350)
Closure	9312786	200
Container Assembly	9312697	350
Container Assembly	9312697	10

4.3.2 Inspection to be performed. See MIL-A-48078 and Table I.

4.3.3 Rejection. See MIL-A-48078.

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4.3.4 Initial qualification of die-press set-ups. Initial die-press set-up qualification shall be accomplished by conventional die maker techniques to check the concentricity of male-female dies. The following procedure shall be employed:

a. Strips of eutectic solder 0.032 inches in thickness are to be placed in the cavity of each die in three places. The dies are then to be closed on the strips. Subsequently the strips are to be removed from the dies and measured with a ball anvil and ball spindle micrometer. Inner diameter, outer diameter, and base thickness measurements are to be obtained and shall be in accordance with the applicable drawing.

b. After the die-press set-ups have been qualified with solder, fifteen each parts shall be processed through each die and measured for inner and outer diameters, and base thickness. All dimensions shall be in accordance with the applicable drawing.

c. All measurements obtained during die-press qualification shall be maintained on file and shall be available for government review. The die-press set-ups shall not be approved for use in production if any part, assembly or solder set-up fails to comply with above requirements.

d. All die-press set-ups shall operate against the established stop and meet an established dwell time.

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TABLE I First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: Parts and Container Assembly for M204, Charge Propelling		SHEET 1 OF 1		DRAWING NUMBER: See Below NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Container, Top (dwg. 9312699)				
	Examination for defects	15/Cav		3.2	4.4.2.2
	Homogeneity of material	350		3.9	Visual (4.5.7)
	Container, Bottom (dwg. 9312700)				
	Examination for defects	15/Cav		3.2	4.4.2.3
	Homogeneity of material	350		3.9	Visual (4.5.7)
	Closure (dwg. 9312786)				
	Examination for defects	15		3.2	4.4.2.4
	Composition 1/	100 2/		3.2	4.5.1
	Stability heat test (134.5°C) 1/	100 2/		3.3	4.5.2
	Container Assembly (dwg. 9312697)				
	Examination for defects	25 3/		3.2	4.4.2.5
	Weight	350 3/		3.2	Gage
	Composition	5 2/		3.2	4.5.1
	Stability heat test (134.5°C)	5 2/		3.3	4.5.2
	Residue test (ballistic)	350 3/		3.5	4.5.3
	Compression strength test	350 3/		3.6	4.5.4
	Fin-boom simulation test	100 3/		3.7	4.5.5
	Water repellancy	350 3/		3.8	4.5.6

NOTES: 1/ An equivalent length of closure material may be used if the closure is manufactured in strip or sheet form. 2/ The same sample may be used for composition and stability heat testing. 3/ The same 350 samples may be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

a. Each lot of container tops and bottoms shall contain parts produced from:

(1) Nitrocellulose from not more than two lots of one lot interfix number from one manufacturer.

(2) Diphenylamine from not more than one lot from one manufacturer.

(3) Acrylic fibers from not more than one lot from one manufacturer.

(4) Polyester fibers from not more than one lot from one manufacturer.

(5) Kraft fibers from not more than one lot from one manufacturer.

(6) Resin from one manufacturer.

(7) Resin additives from one manufacturer.

b. Each lot of container assemblies shall contain parts produced from:

(1) Container tops and container bottoms from not more than one manufacturer and one lot.

(2) Silicone from one manufacturer.

(3) Hexane from one manufacturer.

(4) Acetone from one manufacturer.

(5) MR-23 from one batch from one manufacturer.

(6) Adhesive, cellulose, nitrate base, Type I from one batch from one manufacturer.

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4.4.2 Examination. See MIL-A-48078. Conditioning shall be at 23 ± 2°C and 50 ± 5% R.H. for a minimum of four hours. These conditioning requirements apply to container assembly sample inspections (paragraph 4.4.2.5) that require gaging. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

4.4.2.1 Mold examination and process requirements.

4.4.2.1.1 Qualification for new or rework die-press set-ups. Qualification of new or reworked die-press set–ups shall be in accordance with 4.3.4.

4.4.2.1.2 Process requirements. Homogeneity inspection will be performed prior to assembly of container halves.

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Container, Top				9312699
					NEXT HIGHER ASSEMBLY 9312697
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special A	Homogeneity of material		100%	3.9	Visual (4.5.7)
Major					
101	Thickness of inside wall (average of three places (1/)		0.40%	3.2	Gage
102	Thickness of outside wall (average of three places (1/)		0.40%	3.2	Gage
103	Thickness of top (average of three places) (1/)		0.40%	3.2	Gage
Minor					
201	Height of flat		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.12.1	Visual

NOTES: 1/ In addition, any top with an individual reading greater than .003 inches outside of drawing tolerance shall be considered defective.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9312700
4.4.2.3	Container, Bottom				NEXT HIGHER ASSEMBLY 9312697
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special A	Homogeneity of material		100%	3.9	Visual (4.5.7)
Major 101	Thickness of inside wall (average of three places) (1/)		0.40%	3.2	Gage
102	Thickness of outside wall (average of three places) (1/)		0.40%	3.2	Gage
103	Thickness of bottom (average of three places) (1/)		0.40%	3.2	Gage
Minor 201	Evidence of poor workmanship		0.65%	3.12.1	Visual

NOTES: 1/ In addition, any bottom with an individual reading greater than .003 inches outside of drawing tolerance shall be considered defective.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Closure	1 OF 1	9312786
			NEXT HIGHER ASSEMBLY 9312697

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Composition 1/	100 2/		3.2	4.4.3.1/4.5.1
102	Stability heat test 1/	100 2/		3.3	4.4.3.2/4.5.2
Minor					
201	Length 3/		0.65%	3.2	Gage
202	Width		0.65%	3.2	Gage
203	Thickness		0.65%	3.2	Gage
204	Evidence of poor workmanship		0.65%	3.12.1	Visual

NOTES 1/ An equivalent length of closure material may be used if the closure is manufactured in strip or sheet form.
2/ The same sample may be used for composition and stability heat testing.
3/ This inspection does not apply if the closure is manufactured in strip or sheet form.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Container Assembly	1 OF 2	9312697
			NEXT HIGHER ASSEMBLY 9312698

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Weight		0.25%	3.2	Gage
102	Outside diameter, max. (1/)		0.40%	3.2	Gage
103	Inside diameter, min. (1/)		0.40%	3.2	Gage
104	Distance across open end		0.40%	3.2	Gage
105	Height		0.40%	3.2	Gage
106	Composition	5 2/		3.2	4.4.3.1/4.5.1
107	Stability heat test	5 2/		3.3	4.4.3.2/4.5.2
108	Residue test (ballistic) (3/)	125		3.4	4.5.3
109	Compression strength test		0.40%	3.7	4.5.4
110	Fin-boom simulation test		0.40%	3.8	4.5.5
111	Water repellancy test		0.15%	3.9	4.5.6
112	Incomplete bond between container top and bottom		0.40%	3.2	Visual

NOTES: 1/ Additionally, outside diameter, min. and inside diameter, max. shall be inspected for one lot out of every five lots and during first article in accordance with MIL-STD-105, Level II at an AQL of 0.40%.
2/ The same sample may be used for composition and stability heat testing.
3/ After the first successful production lot, discontinue residue testing.

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Container Assembly (continued)	2 OF 2	9312697
			NEXT HIGHER ASSEMBLY 9312698

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Diameter of loading hole		0.65%	3.2	Gage
202	True position of closure seat		0.65%	3.2	Gage
203	Width of closure tab		0.65%	3.2	Gage
204	True position of loading hole		0.65%	3.2	Gage
205	Evidence of poor workmanship		0.65%	3.12.1	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Packing and Marking for Box, Fiberboard Outer for Charge, Propelling, M204 or Container Assembly	1 OF 1	9313703
			NEXT HIGHER ASSEMBLY 9313727

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Containers incorrectly packed		0.40%	5.1	Visual
102	Marking improper		0.40%	5.1	Visual
103	Improper seal		0.40%	5.1	Visual
104	Orientation stripe on corner of outside box improperly located (with respect to loading holes of charges)*		0.40%	5.1	Visual
105	Folded bag in inner box missing (prior to sealing inner box)		0.40%	5.1	Visual
106	Barrier bag damaged (prior to inserting bagged inner box into outer box)		0.40%	5.1	Visual
107	Barrier bag closed improperly (prior to sealing outer box)		0.40%	5.1	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.12.2	Visual

NOTES:

* Applicable to automated charge loading line only.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Packing for Pallet	1 OF 1	9313727
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Orientation of boxes improper		0.40%	5.2	Visual
102	Strapping loose		0.40%	5.2	Visual
103	Boxes damaged		0.40%	5.2	Visual
Minor	None defined				

NOTES:

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4.4.3 Testing.

4.4.3.1 Composition (see dwg. 9277371 and 9381562). The container assembly and closure samples shall be selected from each lot and tested in accordance with 4.5.1. The container assembly sample shall be selected just prior to application of silicone. The lot shall be rejected if the sample fails to comply with the requirements.

4.4.3.2 Stability heat test (134.5°C). The container assemblies and closure shall be selected from each lot and tested in accordance with 4.5.2. The container assembly sample shall be selected just prior to the application of silicone. The lot shall be rejected if the sample fails to comply with the requirements.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the "Quality Conformance Inspection" tables starting with 4.4.2.2. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and 6.5 herein.

4.5 Test methods and procedures. NOTE: This specification covers sampling and testing of explosive materials/munitions which are potentially hazardous to personnel. It is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials. Inert filled ammunition is to be used, whenever feasible, during test.

4.5.1 Composition.

4.5.1.1 Preparation of samples. Each sample shall be cut into approximately 1/4 inch squares. From each sample, approximately 0.5 gram shall be selected for testing and weighed to the nearest 0.2 milligram (weight A) in an extraction thimble. The same thimble shall be utilized during the extractions in 4. 5.1.3.1 and 4.5.1.3.3. Test is destructive and samples should be removed from the lot.

4.5.1.2 Total volatiles. The samples from 4.5.1.1 shall be dried for three hours at 100 ± 2°C, cooled in a desiccator (1/2 hr. min.), and weighed to the nearest 0.2 mg. (weight W). Calculate the percent total volatiles as follows:

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$$\text{Percent (\%) Total Volatiles} = \frac{A - W}{A} \times 100$$

4.5.1.3 Analytical procedure.

4.5.1.3.1 Solvent extraction. Using a tared, extracoarse, sintered glass thimble, extract each sample from 4.5.1.2, with methylene chloride, for a minimum of 16 hours in a Roweg extractor (MIL-STD-286 Method 104.1.3). Evaporate the extract to dryness, using a current of clean, dry air. This extract is to be used for diphenylamine determination in 4.5.1.3.2. Dry the residues (non-extractable material) in the extraction thimbles by aspirating the thimble and contents until the odor of methylene chloride is not discernible. Dry the thimble plus residue for 1/2 hour minimum at 100 ± 2°C. Cool in a desiccator (1/2 hr. min.). Determine the dry weight of this residue (Weight C) to the nearest 0.2 mg. as the difference between this weight and the weight of the empty thimble. This residue is to be used for the nitrocellulose determination in 4.5.1.3.3.

4.5.1.3.2 Diphenylamine (DPA). To the dried extract from 4.5.1.3.1 add 50 milliliters (ml.) of glacial acetic acid. If necessary, warm slightly to dissolve the extract, then cool to room temperature. Complete the determination of diphenylamine in the solution according to Method 201.1.4 (starting with paragraph 5.5) of MIL-STD-286 except that 25 ml. of methylene chloride is to be added after addition of the potassium iodide solution. When the end point appears to have been reached, (absence of blue color) the flask is stoppered and shaken. If no color appears, the end point has been reached; otherwise, continue titration until the blue color is discharged and shaking produces no further color. Calculate the percent DPA as shown in MIL-STD-286, Method 201.1.4, paragraph 5.13.

4.5.1.3.2.1 Alternate method. Cut 1.0 to 1.1 grams of sample in to 1/4 inches squares or smaller. Dry in microwave moisture analyzer. Transfer the dried sample to a weighing bottle. Cool the sample to room temperature in a desiccator and weigh. Transfer the sample to a square homogenizing beaker. Reweigh the empty weighing bottle. The sample weight is equal to the total weight minus the weight of the empty weighing bottle. Add 50 ml of 90% methanol to the square beaker. Agitate the sample/solvent mixture in the hot ultrasonic bath for 10 minutes. Homogenize the mixture using a Binkmann homogenizer (or equivalent) for 2 minutes or until the sample is completely defiberized. Transfer the mixture to a filtering apparatus fitted with a suitable filter paper.

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Add fresh 90% methanol to the square beaker. Grind this solution in the polytron for 1 minute. Add to the filtering apparatus. Repeat two times. Transfer the filtrate into a 250 ml volumetric flask. Make up to the mark with 90% methanol. Shake the flask well enough to make sure the solution is uniform. Using a pipet, transfer exactly 15 ml of this solution into a 50 ml volumetric flask. Make up to mark with 90% Methanol. Shake well. Measure the absorbance of this solution at 285 nm. Calculate the DPA content according to the following formulas:

$$\text{DPA } \% = \frac{(Y - a)\ 250}{b \times w}$$

Where Y = Absorbance
a = Intercept of calibration curve
b = Slope of calibration curve
w = Sample weight in gms.

4.5.1.3.2.2 Calibration procedure. Dry approximately 1 gram of diphenylamine using the microwave moisture analyzer. Weigh approximately 200 mg to the nearest 0.2 mg. Add this to a 250 ml volumetric flask. Add 90% methanol solvent and dilute to mark. Mix well.

Pipet 5 ml from the 250 ml flask and transfer to a 100 ml volumetric flask. Dilute to mark using the same solvent and mix well. This is the standard stock solution. Volumetrically pipet the following aliquots of the standard solution into 100 ml volumetric flasks.

5 ml Solution in a 100 ml volumetric flask = 0.002 mg/ml
10 ml Solution in a 100 ml volumetric flask = 0.004 mg/ml
15 ml Solution in a 100 ml volumetric flask = 0.006 mg/ml

Dilute all standard solutions to mark and mix well. Zero the spectrophotometer with fresh 90% methanol solvent at 285 nm. Rinse the cell with the first standard twice, then measure the absorbance.

Rinse the same cell with the second standard, then measure its absorbance. Perform the same procedure with the third standard. Input the absorbance and concentration data points into a linear regression program. Obtain the slope, y-intercept and r value (correlation coefficient). If r is less than 0.999, the standard should be rerun.

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4 .5.1.3.3 Nitrocellulose. To provide good heat conduction to the solvent, place a sufficient quantity of glycerin on the flask bottom and on the surface of the heat source to produce a continuous film. Add 10 ml. of ethyl alcohol to the thimble and residue from 4.5.1.3.1 and extract with acetone using the method in 4.5.1.3.1. After extraction dry the residue remaining at 100 ± 2°C for one hour. Cool in a desiccator (1/2 hr. min.), weigh to the nearest 0.2 mg. (Weight D) and save for the acrylic determination in 4.5.1.3.4. Evaporate the acetone extract to dryness and add 25 ml. of n-butyl acetate and 45 ml of glacial acetic acid. Replace the air with a stream of inert gas (CO_2 or N_2). Maintain the inert atmosphere throughout this test. Reflux the solution for five minutes. Add to the flask, via the condenser, 25 ml of 0.7N ferrous ammonium sulfate (prepared per MIL-STD-286, Method 604.1.1) and 10 ml. concentrated hydrochloric acid. Reflux for thirty minutes more to obtain a yellow liquid in which the specimen is completely disintegrated. With the inert gas still running, cool in water bath to room temperature. Add 5 ml. of 20% ammonium thiocyanate solution (MIL-STD-286, Method 706.1). Titrate with a 0.2N titanous chloride solution (MIL-STD-286, Method 601.1). Perform a blank titration on the reagents. Calculate percent NC as follows:

$$\text{Percent (\%) NC} = \frac{(46.69)\ (N)\ (V-B)}{(EXW)} \times 100$$

Where:

V = Volume titanous chloride used for sample, ml.
B = Volume titanous chloride used for blank, ml.
W = Weight of dried sample, gm.
N = Normality of titanous chloride
E = Percent nitrogen in the nitrocellulose

4.5.1.3.3.1 Alternate method. Schulze - Tiemann method for nitrocellulose (NC) determinations.

Assemble the 250 ml Kjeldahl flask, gas evolution apparatus eudio meter/condenser tube, NaOH bath, lead band, heating mantle, transformer and protective shield as shown in Figure 1.

Calibrate the apparatus with dry standard A.C.S. grade KNO_3 Weigh 0.5 - 0.7 g standard samples to 0.1 mg and analyze using the following procedure. The factor is determined using the following equation:

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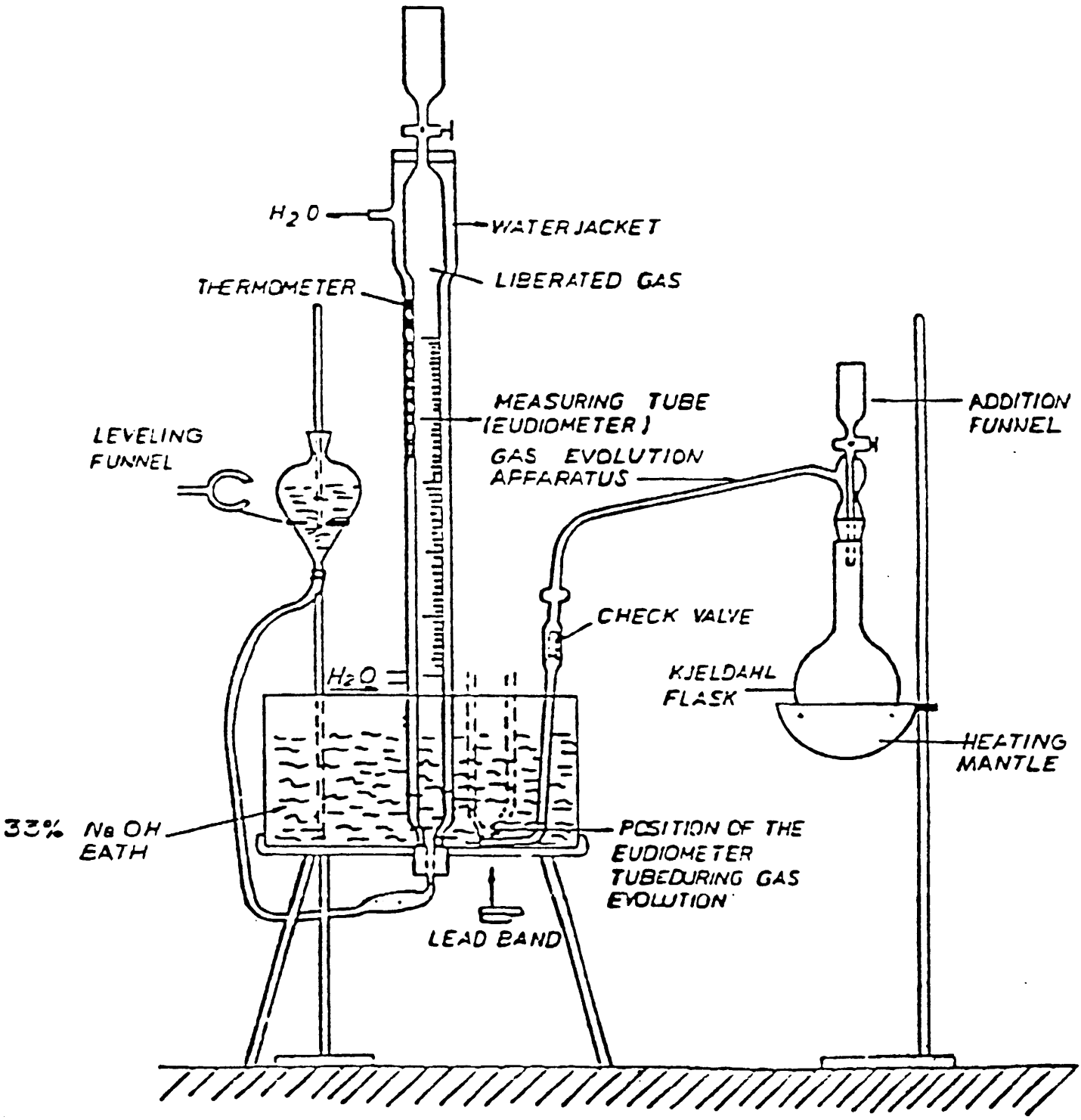


FIGURE 1.
Apparatus for the determination of nitrocellulose by the Schulze-Tiemann method.

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$$F = \frac{616.8\ W\ (273.15 + T)}{V\ (B-C-P-S)}$$

F = Audiometer factor
W = Mass of KNO_3 (g)
T = Temperature of condenser water (°C)
V = Volume of gas in audiometer tube (ml)
B = Barometric pressure (mm Hg)
C = Temperature correction (mm Hg), see paragraph 4.5.1.3.3.2
P = Vapor pressure of water over 33% w/w NaOH at T (mm Hg), see paragraph 4.5.1.3.3.3
S = Local gravity & sea level values corrected to standard values

Prepare samples as follows: Cut a specimen into 1/4 inch or smaller squares. Dry to constant weight using microwave moisture analyzer.

Displace the nitric oxide (NO) from the previous run by aspirating the gases through the drain or venting through a hood . Open the stopcock at the top of the audiometer tube and carefully raise the leveling bulb until all the gas has been expelled. Rest the bulb in its holder and stopper it. Loosen the audiometer tube from the rubber stopper in the bottom of the bath and drain the NaOH in the tube into the tank. Fill the leveling bulb to 90% of its capacity and place the empty audiometer tube back onto the rubber stopper and secure it. Fill the audiometer to the 0.0 ml mark with the leveling bulb, removing all air bubbles, close the stopcock, secure the bulb and stopper it. Replace the 33% w/w NaOH as needed to maintain a constant level in the tank.

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Quantitatively transfer 0.8 - 1.0 g (weighed to ± 0.1 mg) of the dry sample into a 250 ml Kjeldahl flask, add 50 ml deionized water and a few glass beads to prevent bumping. Attach the addition funnel/check valve assembly to the freshly greased joint of the Kjeldahl flask. Attach spring clamps or rubber bands to the ears of the flask and addition funnel. Adjust the gas evolution assembly so that the exit tube is gently resting on the bottom of the NaOH bath. Place the lead band round the exit tube opening.

Open the addition funnel stopcock and turn on the heating mantle. Following the purging of the Kjeldahl flask, close the addition funnel stopcock and purge the rest of the tubing for 5-10 minutes. The system is thoroughly purged when no more bubbles escape into the bath.

Remove the heating mantle from the Kjedahl flask and observe the drawing of the NaOH solution into the exit tubing and the closing of the check valve. Remove the audiometer tube from the rubber stopper. Place it over the open end of the exit tube. Rest the bottom of the audiometer tube on the lead band and secure the audiometer. Refill the leveling bulb to 90% of its volume, stopper the bulb and place it back into its ring.

Place 25 ml of $FeCl_2$ solution into the addition funnel. Open the stopcock slowly and allow most of the $FeCl_2$ to flow into the flask. Retain a few drops in the stopcock to preclude the flow of air into the system. If air is allowed into the system, the analysis is invalid and must be repeated. Place 25 ml of concentrated HCl to the addition funnel and add it to the Kjeldahl flask. Add 10 ml deionized water in a similar fashion. Following the last addition, fill the addition funnel with 30 ml deionized water and tighten the stopcock nut. Place the heating mantle around Kjeldahl flask and heat, for 20-30 minutes or until bubbling has stopped. Remove the heating mantle and check to see that the valve closes. Loosen the stopcock nut on the addition funnel, allow the water to drip into the Kjeldahl, then increase the flow to effectively rinse the walls of the flask, close stopcock before air enters and tighten the nut. Attach the heating mantle, tap the check valve lightly until it opens and continue for 5–10 more minutes.

Following the completion of the reaction, remove the heating mantle and observe the closing of the check valve. Carefully move the audiometer tube from its present position back to the rubber stopper and secure it. Slowly release the vacuum inside the Kjeldahl flask by allowing water and air to enter the flask through the stopcock.

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Tap the check valve to help it open. Allow the gas to equilibrate thermally with the water in the jacket around the eudiometer tube. Wait at least 20 minutes. Measure the volume of gas collected in the audiometer tube by uncorking the leveling bulb and adjusting it so that the top of the fluid in the bulb is at exactly the same height as the top of the fluid in the audiometer tube. Read the volume to the nearest 0.1 ml. Use the following equation to calculate the percent NC in the sample:

$$\% \text{ NC} = \frac{2.246\ V\ (B-C-P-S)\ F}{NW\ (273.15 + T)}$$

V = Volume of gas in audiometer tube (ml)
B = Barometric pressure (mm Hg)
C = Temperature correction (mm Hg), see paragraph 4.5.1.3.3.2
P = Vapor pressure of water over 33% (w/w) NaOH at T (mm Hg), see paragraph 4.5.1.3.3.2
F = Audiometer factor
N = Percent nitrogen in NC (%)
W = Mass of sample (g)
T = Temperature of condenser water (°C)
S = Local gravity & sea level values corrected to standard values

PREPARATION OF $FeCl_2$:

Prepare the solution by adding 1000 g of $FeCl_2.4H_2O$, 1200 ml deionized water and 70 ml of concentrated HCl to a 5 liter round bottom flask. Attach the reflux condenser and reflux for 20 minutes. Cool to room temperature. Slowly add 20 grams of iron fillings or powder to the cool solution. When the gas evolution has stopped, filter quickly without aerating the solution to remove the fines. Use A.C.S. grade chemicals throughout the synthesis.

Prepare the bath fluid by dissolving reagent grade NaOH in deionized water until the concentration is 33% weight to weight.

4.5.1.3.3.2 Barometric pressure temperature correction (variable C). The value for variable C is obtained by using the barometric pressure (mm Hg) and the barometric thermometer temperature (°C) with the following:

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Temperature (°C) Barometric Thermometer	Barometric Pressure in mm Hg		
	740	760	780
14	1.69	1.73	1.78
15	1.81	1.86	1.91
16	1.93	1.98	2.03
17	2.05	2.10	2.16
18	2.17	2.23	2.29
19	2.29	2.35	2.41
20	2.41	2.47	2.54
21	2.53	2.60	2.67
22	2.65	2.72	2.79
23	2.77	2.84	2.92
24	2.89	2.97	3.05
25	3.01	3.09	3.17
26	3.13	3.21	3.30
27	3.25	3.34	3.42
28	3.37	3.46	3.55

4.5.1.3.3.3 Pressure correction value (variable P). The vapor pressure of sodium hydroxide solution as a function of temperature is as follows:

Temperature (°C)	Vapor Pressure of NaOH Solution (mm Hg)
18	11.6
19	12.4
20	13.2
21	14.0
22	14.9
23	15.8
24	16.8
25	17.8

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The calculation of P (actual Pressure) given a temperature T (actual temperature) is determined as follows:

$$P = (T - T_1) \times (P_2 - P_1) + P_1$$

Where T = Actual condenser Water temperature

T_1 = Next lowest tabulated temperature value from T

P_2 = Pressure at next highest tabulated temperature from T

P_1 = Pressure at T_1

4.5.1.3.4 Acrylic.

Reagents

SULFURIC ACID - 93 - 98% H_2SO_4, N_2 Free

MURCURIC OXIDE (H_gO) - Reagent Grade, N_2 Free

POTASSIUM SULFATE OR ANYHYDROUS SODIUM SULFATE - Reagent Grade, N_2 Free

THIOSULFATE SOLUTION - 160 g $Na_2S_2O_3$ $5H_2O$ dissolved in one liter of H_2O

SODIUM HYDROXIDE - 400 g solid NaOH dissolved in H_2O and diluted to one liter

METHYL RED INDICATOR - 1 g methyl red dissolved in 200 ml ethyl alcohol

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METHYLENE BLUE INDICATOR - 1 g methylene blue dissolved in 200 ml ethyl alcohol

HYDROCHLORIC OR SULFURIC ACID STANDARD SOLUTION - 0.1N

BORIC ACID SOLUTION - 4 g H_3BO_4 dissolved in 100 ml H_2O

HENGAR GRANULES

Apparatus

DIGESTION - Use Kjeldahl flask of hard, moderately thick well-annealed glass with total capacity of 500 to 800 ml. Conduct digestion over a safety-approved heating device adjusted to bring 250 ml. H_2O at 25°C to a rolling boil in five minutes.

DISTILLATION - Use a Kjeldhal flask of 500 to 800 ml. capacity fitted with a rubber stopper. The low end of an efficient scrubber bulb or trap is passed through the rubber stopper to prevent mechanical carryover of NaOH during distillation. Use rubber tubing to connect the upper end of the bulb to the condenser tube. Trap the condenser outlet to insure complete absorption of distilled NH_3 into boric acid in the receiver.

Analysis

Quantitatively transfer the residue from 4.5.1.3.3 to a Kjedahl flask, add 3-4 Hengar granules, 0-7 g. of HgO, 15.0g of K_2SO_4 or Na_2SO_4, and 25 ml H_2SO_4 to the flask. Place the flask in an inclined position, and heat gently until frothing ceases. Boil the contents of the flask briskly until the solution clears. Continue boiling for 1/2 hour minimum. Allow the solution to cool to room temperature. Add 200 ml. H20. cool the solution to below 25°C. Add 25 ml. thiosulfate solution and mix to precipitate the mercury. Tilt the flask and add a layer of 100 ml. NaOH solution without agitation. Immediately connect the flask to the distilling bulb on the condenser. With the tip of the condenser immersed in the boric acid in the receiver, rotate the flask to mix the contents thoroughly. Heat the flask until all of the NH_3 has distilled (approximately 250 ml.). Titrate the boric acid solution with the standard acid using a mixed indicator containing four drops of methyl red, and two drops of methylene blue solution. Correct for the reagent blank. Calculate the percent acrylic as follows:

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$$\text{Percent (\%) Acrylic} = \frac{(6.05)\ (V-B)\ (N)}{W} \times 100$$

Where:

V = Volume of standard acid for sample, ml.
B = Volume of standard acid for blank, ml.
N = Normality of standard acid
W = Dried sample weight, g.

4.5.1.3.4.1 Alternate method.

Preparation of samples. Each sample shall be cut into approximately 1/4 inch squares. Cut and dry, in microwave or oven, approximately 2.5 g from each sample and cool in desiccator. From each sample, approximately 1.0 g shall be selected for testing and weighed to the nearest 0.2 mg. Test is destructive and samples should be removed from the lot.

Analytical procedure.

Solvent extraction.

Place the sample in a square beaker and add about 50 mls of absolute methanol. Completely homogenize the sample in methanol for approximately 2 minutes with a polytron homogenizer. Turn the homogenizer to low speed, then with a squirt bottle of technical grade acetone, slowly add the acetone down the shaft of the homogenizer until the level reaches about 1/2 inch from the top of the square beaker. Stir at slow speed for 2 minutes. Set up a vacuum flask with appropriate filter paper and carefully transfer the homogenized sample into the vacuum flask funnel. Immediately wash the shaft of the homogenizer with about 50 mls of acetone and transfer to the vacuum flask funnel. Wash the shaft of the homogenizer again and continue until all of the fibers of the sample have been completely transferred to the vacuum flask funnel. Turn on the vacuum. After the liquid has been completely drained into the flask, wash the sides of the funnel and sample with acetone, assuring that all of the fibers are on the filter paper. Leave the vacuum *on* until the sample is completely dry, about 1 minute. Remove the **vacuum** funnel and carefully remove the filter paper folding it over the sample to insure that no sample is lost.

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Reagents.

SODIUM HYDROXIDE - 400 g solid NaOH dissolved in H_2O and diluted to one liter

THIOSULFATE SOLUTION - 102 g $Na_2S_2O_3$ .5 H_2O dissolved in H_2O and diluted to one liter

SULFURIC ACID - 93-98% H_2SO_4, N_2 Free

MERCURIC OXIDE (HgO) - Reagent grade, N_2 Free

POTASSIUM SULFATE - Reagent grade, N_2 Free

HYDROCHLORIC ACID STANDARD SOLUTION - 0.1N

SODIUM HYDROXIDE STANDARD SOLUTION - 0.1N

BORIC ACID SOLUTION - 4 g H_3BO_4 dissolved in 100 ml H_2O

METHYL RED INDICATOR - 1 g methyl red dissolved in 200 ml ethyl alcohol

METHYLENE BLUE INDICATOR - 1 g methylene blue dissolved in 200 ml ethyl alcohol

POTASSIUM HYDROGEN PHTHALATE - Reagent grade

PHENOLPHTHALEIN INDICATOR - 1 g phenolphthalein dissolved in 200 ml ethyl alcohol

KJEL-FAST CATALYST

Apparatus.

Use a Bucchii distillation unit, a CEM Kjel-FAST digester, or equivalent, a quartz reaction flask, and a distillation adapter.

Digestion.

Turn of the CEM Kjel-FAST digestion unit and water supply (The power switch is on the bottom right corner of the back . The water spigot is behind the unit.)

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The Kjel-FAST unit will prompt:
S-1 SET TIME =
Press 45 ENTER. Then the prompt will be:
S-1 SET POWER =
Press 100 ENTER

There are four stages to the program, the unit will prompt you for the next stage after each is entered. The program is as follows:

S-1 SET TIME = press 45 ENTER
S-1 SET POWER = press 100 ENTER
S-2 SET TIME = press 45 ENTER
S-2 SET POWER = press 100 ENTER
S-3 SET TIME = press 320 ENTER
S-3 SET POWER = press 80 ENTER
S-4 SET TIME = press 100 ENTER
S-4 SET POWER = press 65 ENTER

If a wrong number is pressed, press 0 four times to clear the display. If a wrong number is discover after ENTER is pressed, press ENTER until the display shows SYSTEM READY then press PROG and start from the beginning. If the program is entered correctly, the system status display will read SYSTEM READY and the time remaining display will read 0740. If any problems occur, check the CEM Operation and Service manual for the Kjel-FAST system for more information. With the peristaltic pump in the forward position, lift the switch in the rear of the unit while holding a beaker under the extension tube. Pump Kjel-FAST Catalyst through the system until about 5 mls is pumped through the extension tube. Place the sample and one $K_2SO_4+H_2O$ tablet in the labeled reaction vessel. Add 15 mls of concentrated sulfuric acid to the reaction vessel. Place the reaction vessel with insulator on the vessel pedestal inside the microwave cavity. The vessel will catch the Kjel-FAST Catalyst solution that enters the distribution tube. Press start and run the system to completion for about 8 minutes.

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Distillation.

Before turning on the distillation unit, make sure that the tap water is on. The NaOH reservoir and H_2O reservoir should be checked and filled with a solution of 102 g/l sodium thiosulfate in 10N NaOH (1:5). Depress on/off switch so that the green light comes on. Check to see if proper water circulation occurs. (Large rubber return line should have water flowing through it.) NOTE: If flow rate through this tube is too high, the unit will not operate, it will blow a fuse. Check the operation manual, page 13, instructions 5.1 to insure proper flow rate. Preheat the Buchii unit with about 100 mls of distilled H_2O for five minutes. Wet the distillation system connector with water and place the reaction flask in the unit. Place receiving flask, with 25 mls boric acid (40 g/l) under collection tube, so that the tube is below the liquid level and close the shield. Depress the distillation botton and let unit distill until 100 mls of distillate is collected. Remove collection flask from tube. Let the collection tube drip the remainder of distillate into the boric acid solution. Remove reaction flask from the stand. Pour contents into the nearest acid drain.

Analysis.

Drain the 10 ml buret twice with standardized 0.1N HCl solution. Standardize the pH meter and electrode using pH 4 and H 7 buffer solutions. Titrate the ammonia-boric acid complex with the standard 0.1N HCl to a 4.65 pH. At least two blanks (complete digestion, distillation and pH measurement) are to be performed *every* week. Calculate the percent of acrylic fibers in the specimen as follows:

$$\text{Percent acrylic fibers} = \frac{6.05 \times V \times N \times 100}{W}$$

Where,

V = Volume of standard HCl for sample, ml
N = Normality of standard HCl
W = Weight in grams, of original sample

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4.5.1.3.5 Resin and additives, kraft and polyester staple fiber. Calculate the percent resin and additives, kraft and polyester staple fiber as follows:

% Resin + % additives + % kraft + % polyester staple fiber = 100 - (% NC + % Acrylic + % DPA)

4.5.1.3.6 Methylene chloride extractable. Calculate the total methylene chloride extractable $\frac{(W-C)}{W} \times 100$ and the total acetone extractable $\frac{(C-D)}{W} \times 100$

4.5.2 Stability heat test (134.5°C). Samples shall be tested in accordance with MIL-STD-286, Method 404.1.2, heat tests at 134.5°C. Test is considered destructive and-samples should be removed from lot.

4.5.3 Residue test. The test shall be performed at a Government Proving Ground. Empty containers will be loaded with M10 propellant and sealed in accordance with dwg. 9312698. Loaded containers shall then be assembled to inert-filled M720 cartridges in accordance with the appropriate zones as listed in 4.5.3.1 and 4.5.3.2. Test rounds shall be conditioned for a minimum of eight hours at 70°F plus or minus 5°F and fired within two minutes after removal from conditioning environment. Rounds shall be fired at night from a M224 mortar (standard or ballistic tube) at a firing elevation of 800 roils, at a uniform rate of twelve to fifteen rounds per minute. The time of descent for the round from muzzle to base shall not be more than two seconds. The tube shall be dry swabbed and the elevation checked between the firing of each group.

Observation shall be made for hangups (see 6.8), secondary flame, and flaming debris. The lot shall be accepted if no hangups occur, no rounds exhibit secondary flame at any zone, and no flaming debris reach the ground at zone 4 only.

If a hangup occurs, the test round shall be removed from the tube and an examination shall be conducted of the tube and test round to determine if the hangup was caused by residue. If the hangup was not caused by residue, the test shall be continued. If the hangup was caused by residue, firing of test rounds shall be discontinued. The tube shall be dry swabbed and a group of reference rounds shall be immediately fired.

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The reference round group shall consist of 25 rounds of the same zone, configuration, and conditioning as the test round except that loaded containers from an accepted lot shall be used. The reference group shall be fired after dry swabbing from the same tube used for the test groups. If no hangup occurs in the reference group, the test lot shall be referred.

The test lot shall also be rejected if one or more rounds exhibit secondary flame at any zone or flaming debris reach the ground during zone 4 firing. All other flaming debris results shall be reported for information only. A formal firing report is required (see 6.9).

4.5.3.1 First article. One hundred and seventy five test rounds shall be ballistically tested. These rounds shall consist of seven groups. Each group shall be zone fired in the sequence indicated as follows:

Group	M204 Zone	Test Rounds
1	1	25
2	2	25
3	3	25
4	4	25
5	1	25
6	2	25
7	1	25

4.5.3.2 Production. Fifty test rounds shall be ballistically tested. These rounds shall consist of two groups. Each group hall be zone fired in the sequence indicated as follows:

Group	M204 Zone	Test Rounds
1	1	25
2	4	25

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4.5.4 Compression strength test. The empty container assembly shall be inserted between the plates of a spring compression tester in such a manner that a straight line drawn through the ends of the container would be perpendicular to the plane of the plates. Using 0.3 to 0.5 pounds pressure to hold the container in place, check the perpendicularity of the container to the plates. With a .606 inch precision block held in the opening between the ends of the container assembly, apply a compression load to the container assembly until the container ends make a contact with the precision block. The reading at the initial point of contact with the precision block shall be recorded. Results shall be recorded and evaluated to determine compliance with the requirements. The application of the compression load shall be accomplished within 15 seconds, maximum.

4.5.5 Fin-boom simulation test. The container assemblies shall be tested for flexibility needed during field use. The container assemblies shall be visually examined for any homogeneity defects prior to the test. The defect-free container assemblies shall be positioned on and removed two times from a solid bar of the maximum allowable outside diameter of the fin boom. The "snap" on and off of the boom shall be performed with the container assembly oriented perpendicular to the bar. After removal, the container assemblies shall be visually inspected for cracks, tears, open seams, crushed ends, and fold/creases in the container wall. The government shall establish visual standards prior to production with assistance of the contractor.

4.5.6 Water repellency test. The container assemblies shall be tested for the presence of silicone by brief immersion in water (1 - 2 seconds) and inspection for "beading" or non-wetting. The container is held between the thumb and finger in such a way as to cover the fill hole and permit complete immersion of the container; the assembly shall then be withdrawn and examined. The container shall be rejected if it shows a lack of silicone or incomplete coverage as evidenced by wetting or lack of beading.

NOTE: Container assemblies shall be dried in dry air for at least 24 hours before further use.

4.5.7 Homogeneity of material. Tops and bottoms shall be light table inspected. Any item which fails to conform with the homogeneity requirement shall be classed defective and removed from the lot.

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5. PACKAGING

5.1 Unit packing.

5.1.1 Level A. Not applicable.

5.1.2 Level B. Packing shall be as specified on dwg. 9313703.

5.1.3 Level C. Same as Level B.

5.2 Packing.

5.2.1 Level A. Not applicable.

5.2.2 Level B. Exterior packing shall be as specified on 9313727.

5.2.3 Level C. Same as Level B.

5.3 Marking.

5.3.1 Level A. Not applicable

5.3.2 Level B. Marking shall be as specified on dwg. 9313703 and dwg. 9313727.

5.3.3 Level C. Same as Level B.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The components covered by this specification are intended for use on the 60MM Lightweight Company Mortar System (LWCMS).

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. See MIL-A-48078.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

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c. Orientation stripe on corner of outside box must be specified when required.

6.3 Data requirements. The following Data Item Descriptions (DIDs) must be listed as applicable, on the Contract Data Requirements List (DD Form 1423) when this specification is applied on a contract, in order to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.10	DI-P-1604	Process Description	Use of Contractor's Format

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

(Copies of data item descriptions required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items(s) should be a preproduction sample, a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the Requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

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6.5 Submission of inspection equipment for design approval. See MIL-A-48078. Submit designs as required to Commander, U.S. Army Armament, Munitions and Chemical Command, ATTN: AMSMC-QAT-I (D), Picatinny Arsenal, NJ 07806-5000. This address will be specified in the Contract Data Requirements List DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.6 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as ARRADCOM, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.7 Additives. Additives shall have prior approval by the Contracting Officer. Mold release agents and processing aids are considered additives.

6.8 Hangups. For this specification, the definition of a "hangup" is a round which is dropped into the mortar tube but does not strike the firing pin with enough force to function the primer.

6.9 Submission of formal proving ground firing report. Residue test data shall be sent to Commander, U.S. Army Armament, Munitions and Chemical Command, ATTN: AMSMC-QAT-B (D), Picatinny Arsenal, NJ 07806-5000.

6.10 Proving ground test summary.

Test	Sample Size	Requirement
Residue Test (First Article)	350 Containers (175 test rounds, 25 reference rounds)	Absence of hangup, secondary flame, & flaming debris, zone 4
Residue Test (Production)	125 Containers (50 test rounds)	Absence of hangup, secondary flame, & flaming debris, zone 4

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6.11 Subject term (key word) listing.

Ammunition, mortar
Charge, propelling
Test, residue
Process, slurry
Nitrocellulose

6.12 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project: 1310-A483)

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	MIL-C-48868E (AR)	30 MARCH 1990

3. DOCUMENT TITLE
CHARGE, PROPELLING, M204 PARTS AND CONTAINER ASSEMBLY

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
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US ARMY, ARDEC STANDARDIZATION & SPECIFICATION OFFICE	(1) Commercial (201) 724-6671	(2) AUTOVON 880-6674
c. ADDRESS (Include Zip Code) SMCAR-BAC-S/BLDG 6 PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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